

Riesz-complete extensions realize every Weyl order

Candidate full solution to an open question in arXiv:2510.20249

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Result. The existence of a Riesz-complete non-self-adjoint extension imposes no restriction on Weyl order. For every $\rho \in [0, \infty]$ there is an entire simple symmetric operator with deficiency indices $(1, 1)$, of Weyl order ρ , which has such an extension.

1 The source question and its criterion

For an entire simple symmetric operator T with deficiency indices $(1, 1)$, the source defines the Weyl height $h_T(r)$ and Weyl order

$$\rho_T = \limsup_{r \rightarrow \infty} \frac{\log h_T(r)}{\log r}.$$

At the end of its section on completeness (source PDF page 45), it asks whether the existence of a Riesz-complete non-self-adjoint extension restricts ρ_T .

We use two results proved in the source [1]. If the eigenvalues of a strictly accumulative extension T_∞ are $\lambda_n \in \mathbb{C}_-$ and the contractive Weyl function is the corresponding meromorphic Blaschke product, then

$$T_\infty \text{ is Riesz complete} \iff \inf_n \prod_{m \neq n} \left| \frac{\overline{\lambda_n} - \overline{\lambda_m}}{\overline{\lambda_n} - \lambda_m} \right| > 0. \quad (1)$$

Moreover, writing B for the contractive Weyl function,

$$h_T(r) = \frac{1}{2\pi i} \int_0^r \frac{dt}{t} \int_{-t}^t d \log B(u) + O(1). \quad (2)$$

The criterion is Theorem 6.12 on source PDF page 44 and the height formula is on source PDF page 15. The source also establishes on PDF page 12 that the corresponding entire Nevanlinna curves are realized by entire operators of this kind. Thus it is enough to construct a Blaschke product satisfying (1) and then compute the order in (2).

2 Idea of the proof

Choose the real parts x_n to have whatever radial density is needed for the desired order. This could destroy interpolation if the x_n become crowded. The remedy is to put the poles only an exponentially tiny distance below the real line: if d_n is the nearest-neighbor gap at x_n , use $y_n = 2^{-n}d_n$. For two conjugate poles, the failure of their pseudohyperbolic distance to equal one is then at most $4 \cdot 2^{-n-m}$. Summing logarithms gives a uniform positive lower bound for the full separation product, independently of how dense the real parts become.

Each pole still contributes a fixed positive amount to the phase integral once the integration interval passes its real part. Conversely, its contribution before that point is controlled by y_n/x_n .

Hence the Weyl height has the same power-order as the counting density of the x_n . Three explicit choices of x_n cover positive finite, zero, and infinite order.

3 Uniform interpolation with arbitrary radial density

Let $0 < x_1 < x_2 < \dots$ tend to infinity and set

$$d_n = \min \left\{ 1, \inf_{m \neq n} |x_n - x_m| \right\}, \quad y_n = 2^{-n} d_n, \quad \lambda_n = x_n - i y_n.$$

Since the sequence is discrete, $d_n > 0$. Put $z_n = \overline{\lambda_n}$ and define

$$B(\lambda) = \prod_{n=1}^{\infty} \frac{\lambda - z_n}{\lambda - \overline{z_n}}. \quad (3)$$

The harmless usual unimodular normalizing factors may be inserted. Because $\sum_n y_n < \infty$ and $x_n \rightarrow \infty$, the product converges normally away from its poles. It is a meromorphic inner function on the upper half-plane, with zeros z_n , poles λ_n , and zero exponential factor.

Lemma 1 (Uniform separation). *The sequence (z_n) satisfies*

$$\inf_n \prod_{m \neq n} \left| \frac{z_n - z_m}{z_n - \overline{z_m}} \right| \geq e^{-2}.$$

Consequently the extension with eigenvalues (λ_n) is Riesz complete.

Proof. For $m \neq n$, write

$$r_{nm} = \left| \frac{z_n - z_m}{z_n - \overline{z_m}} \right|.$$

A direct calculation gives

$$r_{nm}^2 = 1 - u_{nm}, \quad u_{nm} = \frac{4y_n y_m}{(x_n - x_m)^2 + (y_n + y_m)^2}.$$

The definition of d_n gives $|x_n - x_m| \geq \max\{d_n, d_m\}$ and therefore

$$0 \leq u_{nm} \leq 4 \cdot 2^{-n-m}.$$

Because $n, m \geq 1$ and $n \neq m$, the right side is at most $1/2$. The elementary inequality $-\frac{1}{2} \log(1-u) \leq u$ for $0 \leq u \leq 1/2$ now yields

$$-\log \prod_{m \neq n} r_{nm} \leq \sum_{m \neq n} u_{nm} < 4 \cdot 2^{-n} \sum_{m=1}^{\infty} 2^{-m} = 4 \cdot 2^{-n} \leq 2.$$

This proves the product bound. The Riesz-completeness conclusion is exactly the source criterion (1). $\square$

4 The Weyl-height estimate

For real u , logarithmically differentiating (3) gives a positive phase density,

$$\frac{1}{i} d \log B(u) = \sum_{n=1}^{\infty} \frac{2y_n}{(u - x_n)^2 + y_n^2} du.$$

Define

$$a_n(t) = \frac{1}{2\pi} \int_{-t}^t \frac{2y_n}{(u - x_n)^2 + y_n^2} du, \quad 0 \leq a_n(t) \leq 1.$$

Tonelli's theorem and (2) give

$$h_T(r) = \sum_n \int_0^r a_n(t) \frac{dt}{t} + O(1). \quad (4)$$

Lemma 2 (Density controls height). *There are absolute constants $c, C > 0$ such that, for all sufficiently large r ,*

$$h_T(r) \geq c \# \{n : x_n \leq r/3\} - C, \quad (5)$$

$$h_T(r) \leq \sum_{x_n \leq 2r} \log \frac{2r}{x_n} + C. \quad (6)$$

Proof. If $x_n \leq r/3$, then $x_n + y_n \leq r/2$ for large r . For every $t \in [r/2, r]$, the interval $[-t, t]$ contains $[x_n - y_n, x_n + y_n]$. Integration on this smaller interval shows $a_n(t) \geq 1/2$. Its contribution to (4) is therefore at least $\frac{1}{2} \log 2$, proving (5).

For the upper bound, if $t \leq x_n/2$, then $|u - x_n| \geq x_n/2$ on $[-t, t]$, whence

$$a_n(t) \leq \frac{8}{\pi} \frac{ty_n}{x_n^2}.$$

For $t > x_n/2$ use only $a_n(t) \leq 1$. The contribution of an index with $x_n \leq 2r$ is at most $Cy_n/x_n + \log(2r/x_n)$; the first terms are summable. If $x_n > 2r$, its contribution is at most Cry_n/x_n^2 , and their sum is bounded (indeed it is $O(r^{-1})$). Substitution in (4) proves (6). $\square$

5 Complete realization of all Weyl orders

Theorem 1 (Full answer to the source question). *For every $\rho \in [0, \infty]$ there is a transcendental entire simple symmetric operator T with deficiency indices $(1, 1)$ and Weyl order $\rho_T = \rho$ which has a Riesz-complete non-self-adjoint extension.*

Proof. Use the construction above. Lemma 1 always gives the Riesz-complete extension T_∞ . Its eigenvalues lie strictly in $\mathbb{C}_-$, so it is non-self-adjoint.

For $0 < \rho < \infty$, take $x_n = n^{1/\rho}$. The counting term in (5) is comparable to r^ρ , while

$$\sum_{n \leq (2r)^\rho} \log \frac{2r}{n^{1/\rho}} = O(r^\rho)$$

by Stirling's formula. Lemma 2 therefore gives $c_\rho r^\rho \leq h_T(r) \leq C_\rho r^\rho$ for large r , and hence $\rho_T = \rho$.

For $\rho = 0$, take $x_n = e^{\sqrt{n}}$. Then

$$c(\log r)^2 \leq h_T(r) \leq C(\log r)^3.$$

Thus $h_T(r)/\log r \rightarrow \infty$, so the operator is transcendental, while $\log h_T(r)/\log r \rightarrow 0$; hence $\rho_T = 0$.

Finally, for $\rho = \infty$, take $x_n = \log(n+1)$. The lower bound gives $h_T(r) \geq ce^{r/3}$ for large r , and consequently $\log h_T(r)/\log r \rightarrow \infty$. This is infinite Weyl order.

All possible values of Weyl order have now been realized, so Riesz completeness imposes no restriction on it. $\square$

6 Verification and novelty scope

The included verifier evaluates finite separation products for all three growth regimes and checks the analytic bound $u_{nm} \leq 4 \cdot 2^{-n-m}$. It also computes truncated phase integrals from (4); these checks are not used in the proof.

A bounded search through 17 August 2026 used the exact paper title, the exact question sentence, and combinations of “Riesz complete”, “Weyl order”, and “interpolating sequence”. It found no later answer; arXiv still lists only version 1 of the source. The result uses the source’s realization, Riesz-basis criterion, and height formula. The interpolating construction and the all-orders conclusion are proved here.

References

- [1] Y. Wang, *Complex analysis of symmetric operators. II: entire operators with deficiency index 1*, arXiv:2510.20249v1 (2025).